Shivam Kumar Verma

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Education

Indian Institute of Information Technology, Nagpur — **November 2022 – May 2026**
B.Tech in Computer Science and Engineering — *Nagpur, Maharashtra*

Experience

poiro.com — **July 2026 – Present**
Fullstack Developer — *TypeScript, Python, FastAPI, Next.js, PostgreSQL, Cloudflare R2*

- **Storage Integrity and Deduplication:** Engineered storage-verified **SHA-256** asset deduplication backed by a **PostgreSQL** uniqueness guarantee and a client-side preflight, preventing duplicate **Cloudflare R2** writes without trusting client-supplied hashes.
- **Creative Asset Manager Backend:** Built the data model, services, and APIs for the asset layer behind AI-generation surfaces; fixed production endpoint failures and reference-integrity defects that hid nested folders, and replaced depth-dependent tree traversal with a **single aggregation query**.
- **Repository Reliability:** Cleared approximately **1,041 Ruff findings across 7 Python trees**, made lint/format checks blocking in CI and pre-push, and added dynamic package discovery so new packages are covered automatically.

klimb.io — **November 2025 – March 2026**
Full Stack Developer Intern — *TypeScript, Node.js, Express.js, PostgreSQL, Redis, AWS*

- **AI Usage Metering:** Designed a **PostgreSQL**-backed usage-metering and credit ledger across **6+ AI features** in a **Node.js**/**Express** backend, replacing binary feature flags with an auditable per-feature usage log for finance.
- **Data Access Optimization:** Rewrote slow **PostgreSQL** queries and restructured recruiter-dashboard data access, reducing p95 latency and memory overhead under concurrent load.
- **AI Interview Platform Backend:** Developed backend services for a **3+ stage** AI interview platform with automated proctoring and strict candidate-level data isolation across concurrent live sessions.

Projects

CodeSM | GitHub | Live | *Node.js, Express.js, BullMQ, Redis, PostgreSQL, AWS S3*

- **Background Job Pipeline:** Architected a **BullMQ + Redis** pipeline that persists each submission in **PostgreSQL** before grading it outside the request lifecycle, keeping the **Express** API responsive and retrying failed jobs automatically.
- **Sandboxed Multi-Language Runtime:** Isolated untrusted code in **6 languages** (C, C++, Java, Go, Python, JavaScript) in Docker with CPU, memory, network, filesystem, timeout, and output limits, reusing containers instead of starting one per submission.
- **Lazy Testcase Processing:** Streamed **AWS S3**-backed test data one case at a time with early exit on the first failure, bounding per-job memory and skipping unnecessary downloads.

Hireflow | GitHub | *Node.js, TypeScript, BullMQ, PostgreSQL, Playwright*

- **Decoupled Ingestion Queues:** Split ingestion from **4 sources** (LinkedIn, Wellfound, YC Work at a Startup, Telegram via GramJS), AI processing, and browser automation onto separate **BullMQ** queues, so a bot-detection block in one stage cannot stall the others.
- **Deduplication and Browser Automation:** Implemented **SHA-256** fingerprints of the normalised title, company, and description so a listing arriving from several sources is stored once in **PostgreSQL**, then automated submission on third-party sites with **Playwright** within each source's rate limits.

Technical Skills

Backend & APIs: Node.js, Express.js, REST APIs, background job processing, WebSockets, FastAPI
Data Systems: PostgreSQL, Redis, Prisma, MongoDB, Cloudflare R2, deduplication pipelines
Infrastructure & Runtimes: AWS (S3, EC2), BullMQ, Docker, Linux, CI/CD, Kubernetes
Languages & Automation: JavaScript, TypeScript, Python, C++, Playwright, React, Next.js, Postman, ESLint

Achievements

- **Competitive Programming: Top 5% globally** on LeetCode (**1800+ rating**, 800+ problems solved) · **3-star** on CodeChef.
- **Tantrafiesta 2024: 3rd rank** at IIIT Nagpur's hackathon.